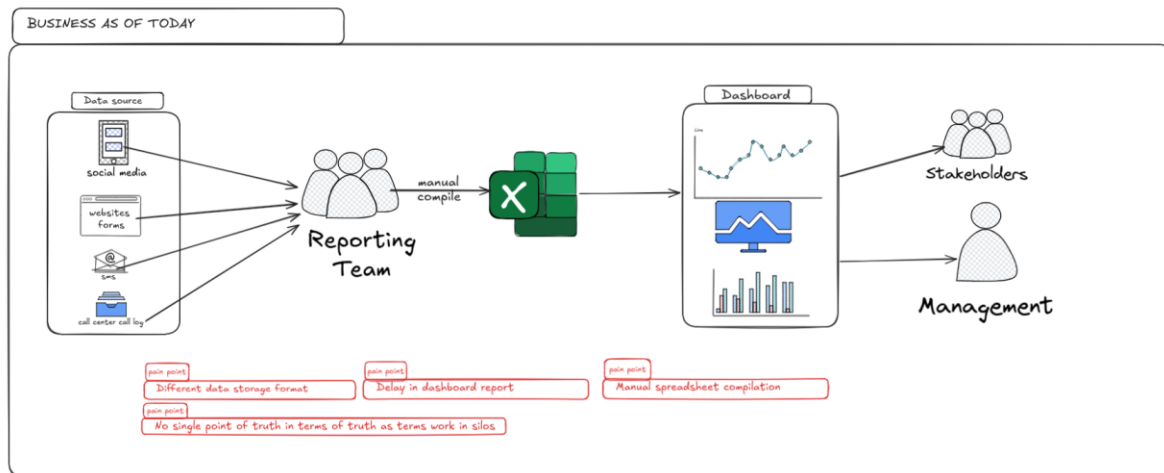


BEEJAN TECHNOLOGIES DATA SOLUTION

1. Current State Assessment



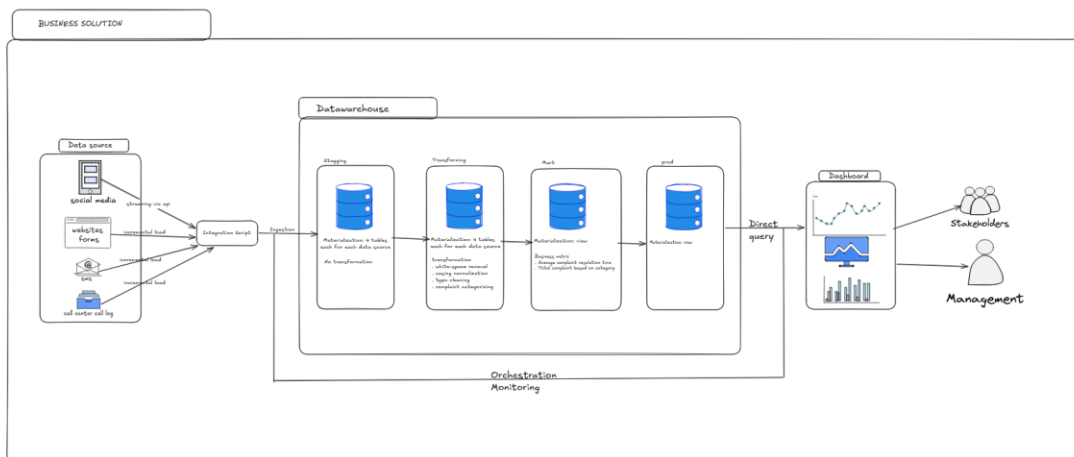
As it stands, the telecom company manages customer complaints through four disconnected channels: **social media**, **SMS**, **website forms**, and **call centre log files**. There is no automated pipeline connecting these sources to the business's reporting layer, only that a reporting team manually pulls data from each platform, consolidates it in spreadsheets, and builds dashboards for stakeholders and management from that manual compilation.

This manual, siloed approach introduces four core pain points:

- **Different data storage formats:** Each source stores complaint data in its own structure (raw text, structured form fields, log entries, social media API payloads), making consolidation error-prone and time-consuming.
- **Delay in dashboard reporting:** Because compilation is manual, dashboards reflect stale data rather than near real-time customer sentiment and complaint volume.
- **Manual spreadsheet compilation:** The reporting team spends significant time on repetitive, low-value data-wrangling work instead of core deep analysis.
- **No single point of truth:** Since each source is processed independently and in silos, stakeholders and management often work from inconsistent or conflicting numbers.

If the above is left unaddressed, these issues slow down the business's ability to detect complaint trends, respond to customer issues proactively, and make timely, data-backed decisions.

2. Proposed Business Solution



The proposed solution replaces the manual workflow with an **automated, end-to-end data pipeline** that takes complaint data from all four sources through ingestion, transformation, storage, and finally into a dashboard layer for direct consumption by stakeholders and management, with orchestration and monitoring running underneath the entire pipeline to guarantee reliability.

At a high level, the architecture covers five stages:

1. **Data Sources:** Social media, website forms, SMS, and call centre logs
2. **Ingestion Strategy:** I'm proposing a hybrid ingestion strategy where we do Streaming ingestion via API for social media platforms, since these sources provide real-time customer engagement and 24-hour incremental batch loads for SMS, call centre logs, and website forms
3. **Data Warehouse (Staging → Transformation → Mart → Prod):** A data warehouse is used as the central storage layer for the pipeline, since the majority of incoming data arrives in CSV and JSON formats.
4. **Orchestration & Monitoring:** The operational layer that schedules, sequences, and observes the whole pipeline
5. **Dashboard / Consumption Layer:** Once data reaches the Prod Layer, it is made available to stakeholders and management via direct query and this helps in eliminating the manual spreadsheet compilation step entirely. Dashboards pull straight from the curated, governed dataset.

3. Summary — How the Solution Maps to the Pain Points

Current Pain Point	How the Proposed Solution Addresses It
Different data storage formats	Standardized in the Transformation Layer before further processing
Delay in dashboard reporting	Streaming ingestion for social media + 24-hour incremental loads for other sources keep data current
Manual spreadsheet compilation	Automated pipeline from ingestion through to the Mart Layer removes manual steps
No single point of truth	Prod Layer serves as the one governed dataset all dashboards and teams query directly

This architecture gives Beejan Technologies a scalable, automated foundation for complaint data, helping to move the business from reactive, manually-compiled reporting to a governed, near real-time pipeline that supports faster decision-making for both stakeholders and management.